

A detailed black and white schematic of a printed circuit board (PCB) layout. The design features a complex network of thick black traces connecting various components. There are numerous circular mounting holes of different sizes distributed across the board. The layout is dense and irregular, with components represented by small geometric shapes and lines. The overall shape is roughly rectangular with several protrusions and indentations.

The diagram illustrates a 900MHz digital frequency synthesizer circuit. Key components include a 900MHz VCO, a 200KHz crystal oscillator, and two 555 timers (IC1 and IC2) for frequency division. The circuit features two sets of 4-bit digital switches (X1, X2) for 'START' and 'STOP' frequency selection. The output is filtered by a 100nH inductor and a 100pF capacitor, and then amplified by a 2N3858 JFET. The circuit is powered by a 9V battery and includes various passive components like resistors and capacitors.

pin 1 being neutral. Live pins 2 are joined by wire to pins 3 and output voltage is taken from pins 1 and 5. There is a provision of button cell CR2032 to hold the clock and programmed times. That means, even if 230V AC is turned off, the clock and programmed times are not disturbed (during mains failure) for 60 to 90 days. When mains power is present, the cell is charged continuously.

two monostable multivibrators for the time delay of five seconds. The start time switch1 is connected to the first monostable multivibrator built around IC1 as shown in Fig. 2.

The real-time clock of time switch1 is set by pressing and holding CLOCK button and adjusting the time by using HOUR, MIN and DAY buttons. If first time duration of 8am to 9am has to be programmed at weekly mode, then 8am is programmed at 1 ON mode and 8.01am is programmed at 1 OFF mode in the first time switch by selecting weekly mode.

The motor is switched off by using the second multivibrator circuit as shown in Fig. 2, in which N/C and common terminals of relay RL2 are connected in series with off switch of the starter.

Real-time clock is set by pressing and holding CLOCK button and adjusting the time by using HOUR, MIN and DAY buttons. Off time, that is, 9am, is programmed at 1 ON mode with weekly day selection by pressing the TIMER button.

Again, by pressing TIMER button, 9.01am is set at 1 OFF mode with weekly day selection in the second time switch. When time 9am is reached, the second time switch provides 230V AC across the primary of step-down transformer X2, and the second full-wave rectifier outputs 12V DC. This voltage goes to the second monostable multivibrator circuit as shown in Fig. 2.

Fig. 5 shows the photograph of a typical starter for a 3-phase electric motor along with the interior assembly of the starter. On the right of the photograph, two push-buttons are shown; the green push-button is used to start the motor and the red push-button is used to stop it. It also has a relay coil. When start switch is pressed momentarily, current flows through the coil, relay strip is pulled towards the iron of the coil and 3-phase voltage gets applied to the motor.

An actual-size, single-side PCB of the 3-phase electric motor is shown in Fig. 6 and its component layout in Fig. 7.

EFY note. Reset the time switch if there is any difficulty in setting time on the time switch. ●



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